روزبه پورجعفریان (۴۰۴۲۰۱۹۸۴)
ریاضی مهندسی پیشرفته
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تاریخ دریافت: ۱۴۰۴/۱۱/۰۳
تاریخ ارسال: ۱۴۰۴/۱۱/۲۳
تمرین شمارهٔ ۵

Q1: Find two term answer for roots.

a) $\epsilon x^3 + x^2 + (2+\epsilon)x + 1 = 0$

→ for double root, add this term.

write x as series: $x = y/\epsilon + x_0 + \sqrt{\epsilon}\,x_1 + \epsilon x_2 + \dots = y/\epsilon + X$

$\epsilon x^3 + x^2 + (2+\epsilon)x + 1 = 0 \Rightarrow \epsilon(y/\epsilon + X)^3 + (y/\epsilon + X)^2 + (2+\epsilon)(y/\epsilon + X) + 1 = 0$

$\Rightarrow \epsilon((y/\epsilon)^3 + 3y^2X/\epsilon^2 + 3yX^2/\epsilon + X^3) + (y^2/\epsilon^2 + 2yX/\epsilon + X^2) + (2+\epsilon)(y/\epsilon + X) + 1 = 0$

$\epsilon x^3:\ y^3/\epsilon^2 + 3y^2X/\epsilon + 3yX^2 + \epsilon X^3$

$x^2:\ y^2/\epsilon^2 + 2yX/\epsilon + X^2$

$(2+\epsilon)x:\ 2y/\epsilon + y + 2X + \epsilon X$

$\Rightarrow (y^3+y^2)/\epsilon^2 + (3y^2X + 2yX + 2y)/\epsilon + (3yX^2 + X^2 + y + 2X + 1) + \underbrace{\epsilon(X^3+X)}_{\text{too small}} = 0$

$\Rightarrow y^3 + y^2 = 0 \Rightarrow y^2(y+1) = 0 \Rightarrow y = 0$ or $y = -1$

• $y=-1$: $3y^2X + 2yX + 2y = 3X - 2X - 2 = X - 2$, $\quad X = 2 + O(\sqrt{\epsilon})$ $\Big\}\ x_0 = 2$

$X = x_0 + \sqrt{\epsilon}\,x_1 + x_2\epsilon + \dots$

$3yX^2 + X^2 + 2X + y + 1 = -3X^2 + X^2 + 2X - 1 + 1 = -2X^2 + 2X = 0$

$X = 2 + \sqrt{\epsilon}\,x_1 + \epsilon x_2 \Rightarrow X^2 = (2 + \sqrt{\epsilon}\,x_1 + \epsilon x_2)^2 = 4 + 4\sqrt{\epsilon}\,x_1 + \epsilon(x_1^2 + 4x_2) + O(\sqrt{\epsilon}^3)$

$\Rightarrow -2X^2 + 2X = -4 + \sqrt{\epsilon}(-6x_1) + \epsilon(-2x_1^2 - 6x_2) + \dots \overset{*}{=} -4 - 6x_2$

$\frac{X-2}{\epsilon} = \frac{2 + \sqrt{\epsilon}\,x_1 + x_2\epsilon - 2}{\epsilon} = \frac{x_1}{\sqrt{\epsilon}} + x_2 + \dots$ → we don't need this term, must be cancel:

$\Rightarrow x_1 = 0^{*}$

$-2X^2 + 2X$, $X = 2 + x_2\epsilon = 2 + O(\epsilon) \Rightarrow X = 2 \Rightarrow -2X^2 + 2X = -2(2)^2 + 2(2) = -4$

$\Rightarrow \frac{X-2}{\epsilon} = \frac{2 + x_2\epsilon - 2}{\epsilon} = x_2$, $\quad \frac{X-2}{\epsilon} - 2X^2 + 2X = 0 \Rightarrow x_2 - 4 = 0 \Rightarrow x_2 = 4$

$\Rightarrow x_f = -1/\epsilon + 2 + 4\epsilon$

• $y=0$: $(3y^2X + 2yX + 2y)/\epsilon = 0/\epsilon = 0$, $\quad 3yX^2 + X^2 + y + 2X + 1 = X^2 + 2X + 1 = (X+1)^2$

$\Rightarrow (X+1)^2 + O(\epsilon) = 0$

$\epsilon^0:\ (X+1)^2 = 0 \Rightarrow (x_0+1)^2 = 0 \Rightarrow x_0 = -1 \Rightarrow x = -1 + \sqrt{\epsilon}\,x_1 + \epsilon x_2 + \dots = -1 + \delta$

$\Rightarrow x + 1 = \sqrt{\epsilon}\,x_1 + \epsilon x_2$, $\quad x^2 = 1 - 2\delta + \delta^2$, $\quad x^3 = -1 + 3\delta - 3\delta^2 + \delta^3$, $\quad (2+\epsilon)x = -2 - \epsilon + 2\delta + \epsilon\delta$

$\Rightarrow \epsilon x^3 + x^2 + (2+\epsilon)x + 1 = (-\epsilon + 3\delta\epsilon - 3\epsilon\delta^2 + \epsilon\delta^3) + (1 - 2\delta + \delta^2) + (-2 - \epsilon + 2\delta + \epsilon\delta) + 1 = 0$

$\delta^0:\ 1 - 2 + 1 - \epsilon - \epsilon = -2\epsilon$, $\quad \delta^1:\ -2\delta + 2\delta + 3\epsilon\delta + \epsilon\delta = 4\epsilon\delta$, $\quad \delta^2:\ \delta^2 - 3\epsilon\delta^2$, $\quad \delta^3:\ \epsilon\delta^3$

$\Rightarrow -2\epsilon + 4\epsilon\delta + \delta^2 - 3\epsilon\delta^2 + \epsilon\delta^3 = 0 \ , \ \delta = \sqrt{\epsilon}x_1 + \epsilon x_2$

$\Rightarrow -2\epsilon + 4\epsilon^{3/2}x_1 + \epsilon x_1^2 + 2\epsilon^{3/2}x_1x_2 + O(\epsilon^2)$

$\epsilon^1: -2\epsilon + \epsilon x_1^2 = 0 \Rightarrow x_1^2 - 2 = 0 \Rightarrow x_1 = \pm\sqrt{2}$

$\epsilon^{3/2}: 4\epsilon^{3/2}x_1 + 2\epsilon^{3/2}x_1x_2 = 0 \Rightarrow 2x_1(2+x_2) = 0 \Rightarrow x_2 = -2$

$\Rightarrow x_{2,3} = -1 \pm \sqrt{2\epsilon} - 2\epsilon + \dots$

roots: $x_1 = -1/\epsilon + 2 + 4\epsilon \ , \ x_{2,3} = -1 \pm \sqrt{2\epsilon} - 2\epsilon$

b) $x^2 + \sqrt{1+\epsilon x} = \exp(1/2 + \epsilon)$

$x = x_0 + \epsilon x_1 + O(\epsilon^2) \ , \ 0 < \epsilon \leq 1$

Expand terms: $\sqrt{1+\epsilon x} = 1 + \epsilon x/2 + O(\epsilon^2) = 1 + \frac{\epsilon}{2}(x_0 + \epsilon x_1) \dots = 1 + \frac{\epsilon x_0}{2} + O(\epsilon^2)$

- $\frac{1}{2+\epsilon} = \frac{1}{2}\frac{1}{1+\epsilon/2} = \frac{1}{2}(1 - \epsilon/2 + O(\epsilon^2)) = \frac{1}{2} - \frac{\epsilon}{4} + O(\epsilon^2) \Rightarrow \exp(1/2+\epsilon)$
$= \exp\left(\frac{1}{2} - \frac{\epsilon}{4} + O(\epsilon^2)\right) = e^{1/2} - \epsilon/4\, e^{1/2} + O(\epsilon^2)$
- $x^2 = (x_0 + x_1\epsilon)^2 = x_0^2 + 2x_0x_1\epsilon + O(\epsilon^2)$

$\Rightarrow x^2 + \sqrt{1+\epsilon x} = x_0^2 + 2\epsilon x_0x_1 + 1 + \frac{\epsilon x_0}{2} + O(\epsilon^2) = (x_0^2+1) + \epsilon\left(2x_0x_1 + \frac{x_0}{2}\right) + O(\epsilon^2)$

$\exp(1/2+\epsilon) = e^{1/2} - \frac{\epsilon}{4}e^{1/2} + O(\epsilon^2)$

$\epsilon^0: x_0^2 + 1 = e^{1/2} \Rightarrow x_0 = \pm\sqrt{e^{1/2}-1}$

$\epsilon^1: 2x_0x_1 + \frac{x_0}{2} = \frac{-1}{4}e^{1/2} \Rightarrow x_1 = \left(\frac{-1}{4}e^{1/2} - \frac{x_0}{2}\right)/2x_0 = \frac{-e^{1/2}}{8x_0} - \frac{1}{4} = \frac{-e^{1/2}}{\pm 8\sqrt{e^{1/2}-1}} - 1/4$

roots: $x_1 = \sqrt{e^{1/2}-1} + \epsilon\left(\frac{-1}{4} - \frac{e^{1/2}}{8\sqrt{e^{1/2}-1}}\right) + O(\epsilon^2) \ , \ x_2 = -\sqrt{e^{1/2}-1} + \epsilon\left(\frac{-1}{4} + \frac{e^{1/2}}{8\sqrt{e^{1/2}-1}}\right)$

c) $xe^{-x} = \epsilon$

$e^{-x} = 1 - x + x^2/2 + O(x^3) \Rightarrow xe^{-x} = x - x^2 + \frac{x^3}{2} + \dots \ , \ x = \epsilon a + \epsilon^2 b + O(\epsilon^3)$

$x^2 = \epsilon^2 a^2 + O(\epsilon^3) \Rightarrow xe^{-x} = (\epsilon a + \epsilon^2 b) - (\epsilon^2 a^2) + O(\epsilon^3) = \epsilon a + \epsilon^2(b - a^2) + O(\epsilon^3)$

$\epsilon: a = 1 \ , \ \epsilon^2: b - a^2 = 0 \Rightarrow b = a^2 = 1 \Rightarrow x_1 = \epsilon + \epsilon^2 + O(\epsilon^3)$

$x e^{-x} = \epsilon \Rightarrow \ln x - x = \ln\epsilon \Rightarrow x - \ln x = -\ln\epsilon = L \Rightarrow x = \ln L + L \ldots$

$\ln(x) = \ln(L + \ln L) = \ln L + \ln\left(1 + \frac{\ln L}{L}\right) = \ln L + O\left(\frac{\ln L}{L}\right)$

$\Rightarrow x - \ln x = L + \ln(L) - \ln(L) = L$

$x_2 = L + \ln L + O(1) = -\ln\epsilon + \ln(-\ln\epsilon) + O(1)$

$$\boxed{\text{roots: } x_1 = \epsilon + \epsilon^2 + O(\epsilon^3),\ x_2 = -\ln(\epsilon) + \ln(-\ln\epsilon) + O(1)}$$

Q2: $\left(\frac{1}{12}L^2 + r^2\theta^2\right)\ddot\theta + r^2\theta\dot\theta^2 + gr\theta\cos\theta = 0$, θ expantion.

$A = \frac{L^2}{12}$, $\alpha = \frac{r^2}{A} = \frac{12r^2}{L^2}$, $\omega_0^2 = \frac{gr}{A} = \frac{12gr}{L^2} \xrightarrow{\times 1/A} (1 + \alpha\theta^2)\ddot\theta + \alpha\theta\dot\theta^2 + \omega_0\theta\cos\theta = 0$

can write: $\alpha\theta^2\ddot\theta + \alpha\theta\dot\theta^2 = \alpha\theta(\theta\ddot\theta + \dot\theta^2) = \alpha\theta\, d(\theta\dot\theta)/dt \Rightarrow \ddot\theta + \omega_0^2\theta\cos\theta + \alpha\theta\frac{d}{dt}(\theta\dot\theta) = 0$ *

assume θ samll: $\cos\theta = 1 - \theta^2/2 + O(\theta^4) \Rightarrow \ddot\theta + \omega_0^2\theta + \alpha\theta\frac{d}{dt}(\theta\dot\theta) - \frac{\omega_0^2}{2}\theta^3 = O(\theta^5)$

let write θ as: $\theta = \phi y(\tau)$, $\tau = \Omega t \Rightarrow \dot\theta = \phi\Omega y'$, $\ddot\theta = \phi\Omega^2 y''$

$\theta\frac{d}{dt}(\theta\dot\theta) = (\phi y)\frac{d}{dt}(\phi y \cdot \phi\Omega y') = \phi^3\Omega^2 y(yy')'$

$\Rightarrow \phi\Omega^2(1 + \alpha\phi^2y^2)y'' + \alpha\phi^3\Omega^2 y(y')^2 + \omega_0^2\phi y\cos(\phi y) = 0 \xrightarrow{\div\phi\Omega^2} (1 + \alpha y^2\phi^2)y'' + \alpha\phi y(y')^2 + \frac{\omega_0^2}{\Omega^2}y\cos(\phi y) = 0$

$\cos(\phi y) = 1 - \frac{1}{2}(\phi y)^2 + O(\phi^4)$, $\Omega = \omega_0(1 + k\phi^2)$

$\Rightarrow \Omega^2 = \omega_0^2(1 + 2k\phi^2) + O(\phi^4)$, $\frac{\omega_0^2}{\Omega^2} = \frac{1}{1 + 2k\phi^2} = 1 - 2k\phi^2 + O(\phi^4)$

$\Rightarrow \frac{\omega_0^2}{\Omega^2}y\cos(\phi y) = (1 - 2k\phi^2)y\left(1 - \frac{\phi^2y^2}{2}\right) + O(\phi^4) = y\left(1 - \phi^2\left(2k + \frac{y^2}{2}\right)\right) + O(\phi^4)$

$= y - \phi^2(2ky + 1/2\, y^3) + O(\phi^4)$

$\Rightarrow$ Substituting: $(y'' + \alpha\phi^2y^2y'') + \alpha\phi^2 y y'^2 + (y - \phi^2(2ky + 1/2\, y^3)) + O(\phi^4) = 0$

$\Rightarrow y'' + y + (\alpha(y^2y'' + yy'^2) - 2ky - 1/2\, y^3) + O(\phi^4) = 0$

no write y as series: $y = y_0 + \phi^2 y_1 + O(\phi^4)$

order 1: $y_0'' + y_0 = 0 \Rightarrow y_0 = \cos\tau$

order 2: $y_1'' + y_1 = -\left[\alpha(y_0^2y_0'' + y_0y_0'^2)_{\text{I}} - 2ky_0 - 1/2\, y_0^3{}_{\text{II}}\right]$

$y_0 = \cos\tau$, $y_0' = -\sin\tau$, $y_0'' = -\cos\tau$, $y_0^2y_0'' = -\cos^3\tau$, $y_0y_0'^2 = \cos\tau\sin^2\tau = \cos\tau - \cos^3\tau$

$\Rightarrow$ Sub I: $y_0^2y_0'' + y_0y_0'^2 = -\cos^3\tau + \cos\tau - \cos^3\tau = \cos\tau - 2\cos^3\tau$, $\cos^3\tau = \frac{1}{4}(3\cos\tau + \cos3\tau)$

$\Rightarrow \cos\tau - 2\cos^3\tau = \cos\tau - 1/3(3\cos\tau + \cos3\tau) = -1/2\cos\tau - 1/2\cos3\tau \xrightarrow{\times\alpha} \frac{-\alpha}{2}\cos\tau - \frac{\alpha}{2}\cos3\tau$

II $-1/2\, y_0^3 = -1/2\cos^3\tau = -1/8(3\cos\tau + \cos3\tau)$

$\Rightarrow y_1'' + y_1 = -\left[\left(\frac{-\alpha}{2}\cos\tau - \frac{\alpha}{2}\cos3\tau\right) - 2k\cos\tau + \left(-1/8(3\cos\tau + \cos3\tau)\right)\right]$

Because frequences should not be same: $-2k-\alpha/2-3/8=0 \Rightarrow K=-\left(\frac{\alpha}{4}+\frac{3}{16}\right)$

$\Rightarrow y_1''+y_1=\left(\frac{\alpha}{2}+\frac{1}{8}\right)\cos 3\tau \Rightarrow y_1=\frac{-1}{8}\left(\frac{\alpha}{2}+\frac{1}{8}\right)\cos 3\tau=-\left(\frac{\alpha}{16}+\frac{1}{64}\right)\cos 3\tau$

$\Omega=\omega_0(1+K\phi^2)=\omega_0\left[1-\left(\frac{\alpha}{4}+\frac{3}{16}\right)\phi^2\right]$

$\Rightarrow y=y_0+\phi^2 y_1\ ,\ \theta(t)=\phi y(\tau)=\phi\left(\cos\tau-\phi^2\left(\frac{\alpha}{16}+\frac{1}{64}\right)\cos 3\tau\right)+O(\phi^5)$

result: $\theta(t)=\phi\cos(\Omega t)-\phi^3\left(\frac{\alpha}{16}+\frac{1}{64}\right)\cos(3\Omega t)+O(\phi^5)$

Q3: $\ddot{u}+\omega_0^2 u=\epsilon u\dot{u}^2\ ,\ u(0)=1,\ \dot{u}(0)=0$

a) straight forward expansion

$u(t,\epsilon)=u_0(t)+\epsilon u_1(t)+\epsilon^2 u_2(t)+\dots\ ,\ \dot{u}=\dot{u}_0+\epsilon\dot{u}_1+\epsilon^2\dot{u}_2\ ,\ \dot{u}^2=\dot{u}_0^2+2\epsilon\dot{u}_0\dot{u}_1+O(\epsilon^2)$

$u\dot{u}^2=(u_0+\epsilon u_1+\dots)(\dot{u}_0^2+2\epsilon\dot{u}_0\dot{u}_1+\dots)=u_0\dot{u}_0^2+\epsilon(u_1\dot{u}_0^2+2u_0\dot{u}_0\dot{u}_1+\dots)$

$\Rightarrow$ Right hand: $\epsilon u\dot{u}^2=\epsilon(u_0\dot{u}_0^2)+O(\epsilon^2)$

$\Rightarrow$ left hand: $\ddot{u}+\omega_0^2 u=(\ddot{u}_0+\omega_0^2 u_0)+\epsilon(\ddot{u}_1+\omega_0^2 u_1)+O(\epsilon^2)$

$\Rightarrow (\ddot{u}_0+\omega_0^2 u_0)+\epsilon(\ddot{u}_1+\omega_0^2 u_1)+O(\epsilon^2)=\epsilon(u_0\dot{u}_0^2)+O(\epsilon^2)$

ϵ^0: $\ddot{u}_0+\omega_0^2 u_0=0\ ,\ u(0)=1,\ \dot{u}(0)=0 \Rightarrow u_0(t)=C_1\cos\omega_0 t+C_2\sin\omega_0 t+u_0(0)=1,\ \dot{u}_0(0)=0$

$\Rightarrow u_0(t)=\cos(\omega_0 t)$

ϵ^1: $\ddot{u}_1+\omega_0^2 u_1=u_0\dot{u}_0^2\ ,\ u_1(0)=0,\ \dot{u}_1(0)=0 \Rightarrow u_0=\cos\omega_0 t\ ,\ \dot{u}_0=-\omega_0\sin\omega_0 t$

$u_0\dot{u}_0^2=\cos(\omega_0 t)\,\omega_0^2\sin^2(\omega_0 t)=\cos(\omega_0 t)\,\omega_0^2\,\frac{1}{2}(1-\cos 2\omega_0 t)=\frac{\omega_0^2}{2}(\cos\omega_0 t-\cos\omega_0 t\cos 2\omega_0 t)$

$\cos\omega_0 t\cos 2\omega_0 t=\frac{1}{2}(\cos 3\omega_0 t+\cos\omega t) \Rightarrow u_0\dot{u}_0^2=\frac{\omega_0^2}{4}(\cos\omega_0 t-\cos 3\omega_0 t)$

$\Rightarrow \ddot{u}_1+\omega_0^2 u_1=\frac{\omega_0^2}{4}(\cos\omega_0 t-\cos 3\omega_0 t)$

$$\begin{cases}\ddot{u}_{1a}+\omega_0^2 u_{1a}=\frac{\omega_0^2}{4}\cos\omega_0 t \\ \ddot{u}_{1b}+\omega_0^2 u_{1b}=-\frac{\omega_0^2}{4}\cos 3\omega_0 t\end{cases}\qquad \begin{array}{l} y_h=C\cos(\omega_0 t)+D\sin(\omega_0 t) \\ u_{1a:p}=kt\sin(\omega_0 t)=y\ ,\ u_{1bp}=A\cos(3\omega_0 t)\end{array}$$

$\dot{y}=k\sin(\omega_0 t)+kt\omega_0\cos(\omega_0 t)\ ,\ \ddot{y}=k\omega_0\cos(\omega_0 t)+k\omega_0\cos(\omega_0 t)-kt\omega_0^2\sin(\omega_0 t)$

$=2k\omega_0\cos\omega_0 t-kt\omega_0^2\sin(\omega_0 t)$

$\Rightarrow \ddot{y}+\omega_0^2 y=2k\omega_0\cos\omega_0 t \Rightarrow 2k\omega_0=\frac{\omega_0^2}{4} \Rightarrow k=\frac{\omega_0}{8} \Rightarrow u_{1ap}=\frac{\omega_0}{8}t\sin(\omega_0 t)$

$\dot{u}_{1bp}=-3\omega_0 A\sin(3\omega_0 t)\ ,\ \ddot{u}_{1bp}=(-3\omega_0)^2 A\cos(3\omega_0 t)=-9\omega_0^2 A\cos(3\omega_0 t)$

$\Rightarrow \ddot{u}_{1bp}+\omega_0^2 u_{bp}=(-9\omega_0^2 A+\omega_0^2 A)\cos(3\omega_0 t)=-8\omega_0^2 A\cos(3\omega_0 t)$

$\Rightarrow -8\omega_0^2 A\cos(3\omega_0 t) = \frac{-\omega_0^2}{4}\cos(3\omega_0 t) \Rightarrow A = 1/32 \Rightarrow u_{1bp} = \frac{1}{32}\cos(3\omega_0 t)$

$\Rightarrow u_1 = C\cos\omega_0 t + D\sin\omega_0 t + \frac{\omega_0}{8} t\sin(\omega_0 t) + \frac{1}{32}\cos(3\omega_0 t) \qquad u_1(0)=0,\ \dot u_1(0)=0$

$\Rightarrow C+0+0+1/32 = 0 \Rightarrow C = -1/32$

$\dot u_1 = -C\omega_0\sin\omega_0 t + D\omega_0\cos\omega_0 t + \frac{\omega_0}{8}(\sin\omega_0 t + t\omega_0\cos\omega_0 t) - \frac{3\omega_0}{32}\sin 3\omega_0 t = 0 \Rightarrow D = 0$

$\Rightarrow u_1 = \frac{1}{32}(\cos(3\omega_0 t) - \cos\omega_0 t) + \frac{\omega_0}{8} t\sin(\omega_0 t)$

secular term grow for big t.

answer: $u(t) = \cos(\omega_0 t) + \epsilon\left(\frac{1}{32}(\cos(3\omega_0 t) - \cos\omega_0 t) + \frac{\omega_0}{8} t\sin\omega_0 t\right) + O(\epsilon^2)$

b) averaging to determinate a first_order uniform expansion.

Amplitude phase: $u(t) = a(t)\cos\theta(t)$, $\theta(t) = \omega_0 t + \beta(t)$

$a(t), \beta(t)$ vary slow: $\dot a, \dot\beta = O(\epsilon)^* \Rightarrow \dot u = \dot a\cos\theta - a\sin\theta\dot\theta = \dot a\cos\theta - a\sin\theta(\omega_0+\dot\beta)$

$\overset{*}{\Rightarrow} \dot u = -a\omega_0\sin\theta, \quad \ddot u = -\omega_0\dot a\sin\theta - a\omega_0\cos\theta\dot\theta, \quad \dot\theta = \omega_0+\dot\beta$

$\Rightarrow \ddot u = -\omega_0\dot a\sin\theta - a\omega_0\cos\theta(\omega_0+\dot\beta) = -a\omega_0^2\cos\theta - \omega_0\dot a\sin\theta - a\omega_0\dot\beta\cos\theta$

Sub. in $\ddot u + u\omega_0^2 = \epsilon u\dot u^2 \Rightarrow -\omega_0\dot a\sin\theta - a\omega_0\dot\beta\cos\omega t = \epsilon u\dot u^2 \Rightarrow \dot a\sin\theta + a\dot\beta\cos\theta = \frac{\epsilon}{\omega_0}u\dot u^2$

$u = a\cos\theta \Rightarrow \dot u = \dot a\cos\theta - a\sin\theta\dot\theta = \dot a\cos\theta - a\sin\theta(\omega_0+\dot\beta) = -a\omega_0\sin\theta$

$$\Rightarrow \begin{cases} \dot a\cos\theta - a\dot\beta\sin\theta = 0 \\ \dot a\sin\theta + a\dot\beta\cos\theta = \frac{-\epsilon}{\omega_0}u\dot u^2 \end{cases} \Rightarrow \dot a = \frac{-\epsilon}{\omega_0}(u\dot u^2)\sin\theta, \quad \dot\beta = \frac{-\epsilon}{a\omega_0}(u\dot u^2)\cos\theta$$

$$u\dot u^2 = (a\cos\theta)(a^2\omega_0^2\sin^2\theta) = a^3\omega_0^2\cos\theta\sin^2\theta \Rightarrow \begin{cases} \dot a = -\epsilon a^3\omega_0\cos\theta\sin^3\theta \\ \dot\beta = -\epsilon a^2\omega_0\cos^2\theta\sin^2\theta \end{cases}$$

$\langle\cos\theta\sin^3\theta\rangle = \frac{1}{2\pi}\int_0^{2\pi}\cos\theta\sin^3\theta\, d\theta$, $y=\sin\theta$, $dy=\cos\theta\, d\theta \Rightarrow \int y^3 dy = 0 \Rightarrow \dot a = 0$

$\langle\cos^2\theta\sin^2\theta\rangle = \langle\frac14\sin^2 2\theta\rangle = \frac{1}{2\pi}\int_0^{2\pi}\frac14\sin^2 2\theta\, d\theta = \frac14\times\frac12 = 1/8 \Rightarrow \dot\beta = -\epsilon a^2\omega_0/8 = \frac{-\epsilon a^2\omega_0}{8}$

IC: @ $t=0 \Rightarrow u(0) = a(0)\cos\beta(0) = 1$, $\dot u(0) = -a(0)\sin\beta(0) = 0$, $a(0)=1$, $\beta(0)=0$

$\Rightarrow a(t)=1$, $\beta(t) = \beta(0) - \frac{\epsilon a^2\omega_0}{8}t = -\frac{\epsilon\omega_0}{8}t \Rightarrow \theta(t) = \omega_0 t + \beta(t) = \omega_0 t - \frac{\epsilon\omega_0}{8}t$

$\Rightarrow$ first order uniform: $u(t) \simeq \cos(\omega_0(1-\epsilon/8)t)$

wave for correction: $u(t) = \cos(\Omega t) + \frac{\epsilon}{32}(\cos(3\Omega t) - \cos\Omega t) \qquad \Omega = \omega_0(1-\epsilon/8)$

Q4: $(x+\epsilon y)y' - \frac{1}{2}y = 1+x^2$, First order expansion small ϵ

a) nonuniformity region

$(x+\epsilon y)y' - \frac{1}{2}y = 1+x^2 \rightarrow y' = (1+x^2+\frac{1}{2}y)/(x+\epsilon y)$, $y = y_0 + \epsilon y_1 + \dots$. $0 < \epsilon \ll 1$

ϵy is small correction to x in coef of y': $|\epsilon y| \ll |x|$

$y_0 = O(1) \rightarrow \epsilon y_0 = O(\epsilon) \rightarrow |x| \geqslant \epsilon$

perturbation become non-uniform when small correction ϵy is no longer small:

$|x| \sim |\epsilon y| \rightarrow x = O(\epsilon)$

$\rightarrow x+\epsilon y$ can be smaller and y' destroying outer expansion. $x+\epsilon y \simeq 0$

using $y \simeq y_0 = O(1)$, this occurs inside $O(\epsilon)$ neighbor of $x=0$

answer: $O(\epsilon)$ neighbor $x=0$ where $x+\epsilon y \simeq 0$

b) compare exact solution with approximate solution.

initial condition: for $x=0$, $\epsilon \rightarrow 0$: $y(0) = -2$

$y(x,\epsilon) = y_0(x) + \epsilon y_1(x) + O(\epsilon^2)$

ϵ^0: $xy_0' - \frac{1}{2}y_0 = 1+x^2 \rightarrow y_0' - \frac{1}{2x}y_0 = \frac{1}{x} + x$, ODE: $y_0' + P(x)y_0 = Q(x)$, $P(x) = \frac{-1}{2x}$

$\mu(x) = \exp\int P(x)dx = \exp\int \frac{-1}{2x}dx = |x|^{-1/2}$, $x>0 \Rightarrow \mu = x^{-1/2}$

$\rightarrow x^{-1/2}y_0' - \frac{1}{2x}x^{-1/2}y_0 = x^{-1/2}(\frac{1}{x}+x) \rightarrow \frac{d}{dx}(x^{-1/2}y_0) = x^{-1/2}(\frac{1}{x}+x) = x^{-3/2} + x^{1/2}$

$\int \rightarrow x^{-1/2}y_0 = \int(x^{-3/2}+x^{1/2})dx + C_0 \rightarrow x^{-1/2}y_0 = -2x^{-1/2} + \frac{2}{3}x^{3/2} + C_0 \Rightarrow y_0 = -2 + \frac{2}{3}x^2 + C_0\sqrt{x}$

by initial condition: $y_0(x) = -2 + \frac{2}{3}x^2$

$\epsilon^{(1)}$: $xy_1' - \frac{1}{2}y_1 = -y_0y_0' \rightarrow y_0' = \frac{4}{3}x \rightarrow -y_0y_0' = -(-2+\frac{2}{3}x^2)(4/3x) = \frac{8}{3}x - \frac{8}{9}x^3$

$\rightarrow xy_1' - \frac{1}{2}y_1 = \frac{8}{3}x - \frac{8}{9}x^3 \rightarrow y_1' - \frac{1}{2x}y_1 = \frac{8}{3} - \frac{8}{9}x^2$, $y_1' + P(x)y_1 = Q(x)$, $P(x) = \frac{-1}{2x}$, $Q(x) = \frac{8}{3} - \frac{8}{9}x^2$

$\mu = \exp(\int P(x)dx) = \exp\int \frac{-1}{2x}dx = x^{-1/2}$

$\Rightarrow x^{-1/2}y_1' - \frac{1}{2x}x^{-1/2}y_1 = x^{-1/2}(\frac{8}{3} - \frac{8}{9}x^2) \rightarrow \frac{d}{dx}(x^{-1/2}y_1) = \frac{8}{3}x^{-1/2} - \frac{8}{9}x^{3/2}$

$\int \rightarrow x^{-1/2}y_1 = \frac{16}{3}x^{1/2} - \frac{16}{45}x^{5/2} + C_1 \rightarrow y_1 = C_1\sqrt{x} + \frac{16}{3}x - \frac{16}{45}x^3$, initial cond: $C_1 = 0$

$\rightarrow y_{app} = -2 + 2/3x^2 + \epsilon(\frac{16}{3}x - \frac{16}{45}x^3)$

PAPCO

Numerical: $y' = (1+x^2+y/2)/(x+\epsilon y)$, $y(0)=-2$; use a code for it (attached)

$\epsilon=0.05$, $y(0)=-2$ $x\in[0,2]$

x	y_{num}	y_{app}	Error
0	-2	-2	0
0.2	-1.92889	-1.92012	-0.00877
0.5	-1.67676	-1.70213	0.02537
1	-1.04075	-1.08475	0.04401
1.5	-0.11594	-0.16021	0.04428
2.0	1.09385	1.05778	0.0367

max error: 0.0526

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c) eliminate non-uniformity of response

$y_{out} = -2 + 2/3x^2 + \epsilon(16/3x - 16/45x^3)$, for inner, we use scalling :

$X = x/\epsilon \Rightarrow x = X\epsilon$, $y(x) = Y(X) \Rightarrow y'(x) = \frac{1}{\epsilon} Y_X$

$\Rightarrow (\epsilon X + \epsilon Y)(\frac{1}{\epsilon} Y_X) - \frac{1}{2} Y = 1 + \epsilon^2 X^2 \Rightarrow (X+Y) Y_X - \frac{1}{2} Y = 1 + \epsilon^2 Y^2$

leading order $\epsilon \to 0$: $(X+Y_0) Y_{0X} - \frac{1}{2} Y = 1 \Rightarrow Y_0(X) = -2$ ($Y_{0X} = 0$; $Y_0 = cte$)

inner: $y \simeq -2$, outer: $x \to 0$: $y_{out}(x) \Rightarrow -2$

overlap region: $\epsilon \ll x \ll 1$: $X \to \infty$ while $x \to 0$ $\begin{cases} \text{outer: } y_{out} \sim -2 + O(x) \\ \text{inner : } Y_0 \sim -2 \end{cases}$

$\Rightarrow y_{overlap} = -2$

$y_{compose} = y_{out} + y_{in} - y_{overlap} = y_{out} + Y_0 + 2 = y_{out} - 2 + 2 = y_{out}$

$\Rightarrow y_{compose} = y_{out} \Rightarrow y_{uniform} = -2 + 2/3x^2 + \epsilon(\frac{16}{3}x - \frac{16}{45}x^3)$

Here is no differnce with y_{out} and previous answers, because base on our definition of BC, $C_0 = C_1 = 0$, and $\sqrt{x}$ disappeared and inner scalling now is match to inner solution $Y_0 = -2$.

Q5: $x'' + \epsilon x'(1-x^2) + x = 0$, $x(0)=1$, $x'(0)=0$, priodic solution

$\tau = \omega t$, $\omega = 1 + \epsilon\omega_1 + \epsilon^2\omega_2 + \dots$, $x(\tau) = x_0(\tau) + \epsilon x_1(\tau) + \epsilon^2 x_2(\tau) + \dots$

$\frac{d}{dt} = \omega \frac{d}{d\tau}$, $\frac{d^2}{dt^2} = \omega^2 \frac{d}{dt^2}$ $\Rightarrow$ $\omega^2 x_{\tau\tau} + \epsilon\omega x_\tau (1-x^2) + x = 0$

$\Rightarrow (1 + 2\epsilon\omega_1 + \epsilon^2(\omega_1^2 + 2\omega_2) + \dots)(x_{0\tau\tau} + \epsilon x_{1\tau\tau} + \epsilon^2 x_{2\tau\tau} + \dots) + \epsilon(1 + \epsilon\omega_1 + \epsilon^2\omega_2 + \dots)$

$(x_{0\tau} + \epsilon x_{1\tau} + \epsilon^2 x_{2\tau} + \dots)(1 - (x_0 + \epsilon x_1 + \dots)^2) + (x_0 + \epsilon x_1 + \epsilon^2 x_2 + \dots) = 0$

ϵ^0: $x_{0\tau\tau} + x_0 = 0$ $\Rightarrow$ we know: $x_0(\tau) = A\cos(\tau) + B\sin(\tau) = R\cos(\tau - \phi)$, change phase is like as change initial time. $\Rightarrow x_0(\tau) = R\cos\tau \xrightarrow{\text{rename } R} x_0(\tau) = A\cos\tau$ *

ϵ^1: $x_{1\tau\tau} + x_1 = -2\omega_1 x_{0\tau\tau} - x_{0\tau}(1 - x_0^2) \overset{*}{=} 2\omega_1 A\cos\tau + A\sin\tau - A^3\sin\tau\cos^2\tau$

$\cos^2\tau = (1+\cos2\tau)/2$, $\sin\tau\cos^2\tau = (\sin3\tau - \sin\tau)/2$

$\Rightarrow x_{1\tau\tau} + x_1 = 2\omega_1 A\cos\tau + A(1 - A^2/4)\sin\tau - (A^3/4)\sin3\tau$

resonant term must be absent: $\sin\tau$, $\cos\tau$,

$2\omega_1 A = 0 \Rightarrow \omega_1 = 0$, $A(1 - A^2/4) = 0 \Rightarrow A = 0$ ✗, $A = 4$ ✓ $\Rightarrow \omega_1 = 0$, $A = 4$

$\Rightarrow x_{1\tau\tau} + x_1 = -2\sin3\tau$, $x_{1p} = K\sin3\tau$, $x_{1p_\tau} = 3K\cos3\tau$, $x_{1p_{\tau\tau}} = -9K\sin\tau$

$\Rightarrow -8K\sin3\tau = -2\sin3\tau \Rightarrow K = 4 \Rightarrow x_{1p} = 1/4\sin3\tau \Rightarrow x_1 = 1/4\sin3\tau$

ϵ^2: $x_{2\tau\tau} + x_2 = -2\omega_2 x_{0\tau\tau} - \left[x_{1\tau}(1 - x_0^2) - 2x_0 x_1 x_{0\tau}\right] = (4\omega_2 + \frac{1}{4})\cos\tau + \frac{3}{4}\cos3\tau + \frac{5}{4}\cos5\tau$

absent resonant term: $4\omega + 1/4 = 0 \Rightarrow \omega_2 = -1/16 \Rightarrow \omega = 1 - \epsilon^2/16 + O(\epsilon^3)$

$\Rightarrow x_{2\tau\tau} + x_2 = 3/4\cos3\tau + 5/4\cos5\tau$, guess: $x_2(\tau) = a\cos3\tau + b\sin5\tau$

$x_{2\tau\tau} + x_2 = (-9a + a)\cos3\tau + (-25b + b)\cos5\tau = -8a\cos3\tau - 24b\cos5\tau$

$\Rightarrow -8a = 3/4 \Rightarrow a = -3/32$, $-24b = 5/4 \Rightarrow b = -5/96 \Rightarrow x_2(\tau) = \frac{-3}{32}\cos3\tau - \frac{5}{96}\cos5\tau$

answer: $x(t) = 2\cos\tau + \epsilon\frac{1}{4}\sin(3\tau) + \epsilon^2\left(\frac{-3}{32}\cos(3\tau) - \frac{5}{96}\cos(5\tau)\right) + O(\epsilon^3)$

$\tau = (1 - \frac{\epsilon^2}{16})t$, $T = \frac{2\pi}{\omega} = 2\pi\left(1 + \frac{\epsilon^2}{16}\right) + O(\epsilon^3)$